

The Human Eye and the Colourful World — MCQ Worksheet

Science • Chapter 10 • 25 Questions, 1 mark each — Total: 25 marks

Instructions: Choose the correct option (A / B / C / D) for each question. Each question carries 1 mark. There is no negative marking. Suggested time: 30 minutes.

Q1. The image on the retina is:

- (A) Virtual and erect
- (B) Real and inverted
- (C) Real and erect
- (D) Virtual and inverted

Q2. The ability of the eye lens to change focal length is called:

- (A) Persistence of vision
- (B) Power of accommodation
- (C) Persistence of focus
- (D) Refraction

Q3. The least distance of distinct vision for a normal adult is approximately:

- (A) 10 cm
- (B) 25 cm
- (C) 50 cm
- (D) 100 cm

Q4. The far point of a normal eye is at:

- (A) 25 cm
- (B) 1 m
- (C) Infinity
- (D) 6 m

Q5. The size of the pupil is controlled by the:

- (A) Cornea
- (B) Iris
- (C) Retina
- (D) Lens

Q6. Photoreceptors that respond to colour are called:

- (A) Rods
- (B) Cones
- (C) Neurons
- (D) Muscles

Q7. In myopia (short-sightedness), the image is formed:

- (A) Behind the retina
- (B) On the retina
- (C) In front of the retina
- (D) On the cornea

Q8. Myopia is corrected using a:

- (A) Concave lens
- (B) Convex lens
- (C) Bifocal lens
- (D) Cylindrical lens

Q9. Hypermetropia (long-sightedness) is corrected using:

- (A) Concave lens
- (B) Convex lens
- (C) Plane lens
- (D) Prism only

Q10. The defect corrected by bifocal lenses is:

- (A) Myopia
- (B) Hypermetropia
- (C) Presbyopia
- (D) Cataract

Q11. Cataract is treated by:

- (A) Wearing glasses
- (B) Surgery to replace the lens
- (C) Eye drops only
- (D) Antibiotics

Q12. Splitting of white light into seven colours through a prism is called:

- (A) Reflection
- (B) Refraction
- (C) Dispersion
- (D) Diffraction

Q13. VIBGYOR represents:

- (A) Names of mirrors
- (B) Seven colours of the spectrum
- (C) Eye defects
- (D) Brand of lenses

Q14. Which colour bends the most while passing through a prism?

- (A) Red
- (B) Yellow
- (C) Green
- (D) Violet

Q15. Which colour bends the least while passing through a prism?

- (A) Violet
- (C) Blue

- (B) Indigo
- (D) Red

Q16. Newton recombined the spectrum into white light by using:

- (A) A concave mirror
- (B) A second prism
- (C) A convex lens
- (D) A black surface

Q17. Stars twinkle because of:

- (A) Their own changing brightness
- (B) Atmospheric refraction
- (C) Reflection
- (D) Scattering only

Q18. Planets do not twinkle because:

- (A) They are very small
- (B) They are nearer; appear as extended sources
- (C) They have no light
- (D) They are stationary

Q19. Why does the sun appear red at sunrise/sunset?

- (A) Light passes through more atmosphere; blue is scattered out, only red reaches us
- (B) Sun is hotter then
- (C) Sun is closer at sunset
- (D) Earth tilts back

Q20. The blue colour of the sky is due to:

- (A) Reflection of blue light by oceans
- (B) Scattering of light by air molecules; shorter wavelengths scatter more
- (C) Refraction of blue light
- (D) Dispersion in clouds

Q21. The phenomenon that makes the path of a beam visible in mist is called:

- (A) Dispersion
- (B) Tyndall effect
- (C) Refraction
- (D) Reflection

Q22. The colour of danger signals is red because:

- (A) Red is brighter
- (B) Red light penetrates fog and dust the most
- (C) Red is rare
- (D) Tradition only

Q23. A rainbow is caused by:

- (A) Reflection only
- (B) Dispersion + total internal reflection in water droplets
- (C) Refraction only
- (D) Scattering only

Q24. In rods and cones, rods are responsible for:

- (A) Colour vision
- (B) Bright light vision
- (C) Vision in dim light
- (D) Tear production

Q25. The function of the ciliary muscles is to:

- (A) Move the eyeball
- (B) Change the focal length of the eye lens
- (C) Produce tears
- (D) Filter light